A SPONSOR-ORIENTED RELATIONAL DATABASE SYSTEM FOR DATA-DRIVEN `FOOTBALL INVESTMENT AND REVENUE OPTIMIZATION

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Sponsorship:

FIFA Premier League LaLiga BUNDESLIGA LIGUE 1

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Agenda

1 BACKGROUND

2 SYSTEM ANALYSIS

3 DATA NORMALIZATION

4 DATABASE DESIGN

5 SQL ANALYSIS

6 CONCLUSION

A high-integrity database to transform unreliable football database into a strategic asset for sponsorship investment

SITUATION

Commercial Ecosystem of the "Big Five" Leagues

~8.7B USD
total sponsorship in the top 5 European football leagues annually

>39%
total revenue was generated via sponsorship

Major sponsors are under increasing pressure to:

- Move away from subjective reputation-based investments
- Justify their rising costs **with objective, data-driven evidence.**

COMPLICATION

The Need for Deeper, Risk-Adjusted Performance Metrics

1. Traditional "headline metrics", such as league position or social media following, are **insufficient** for assessing modern sponsorship value

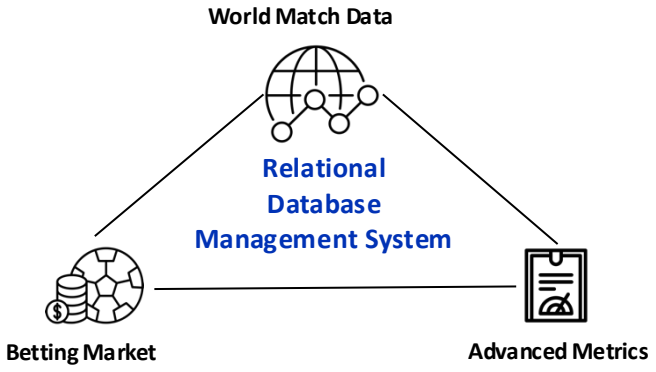
2. Overlook critical on-pitch dynamics that directly impact brand visibility and risk, such as **attacking intensity, consistency, and player discipline**

Without advanced analytics, brands cannot distinguish between a high-performing asset and a high-risk one, leaving a gap between raw football statistics and the commercial outcomes (like screen time and engagement) that investors care about.

RESOLUTION

Relational Database for Sponsorship Intelligence

The project resolves this gap by designing and implementing:



THIS SYSTEM ENABLES:

- Simulate investment screening via complex SQL queries.
- Normalizes data to ensure statistical fairness
- Identify low-risk, high-impact player ambassadors
- Transform raw data into actionable sponsorship insights.

Current CSV-based Kaggle's raw data is fragmented, manual, and unsuitable for sponsor-grade analytics, blocking sponsorship-relevant insights

A classic unscalable flat-file system rather than a true database:

- Isolated table with **no enforced relationships.**
- **No guarantee of identifiers consistency** across files.
- **Untyped text -** no guarantee that data are valid or comparable.
- **Repeated business concepts** instead of shared entities.

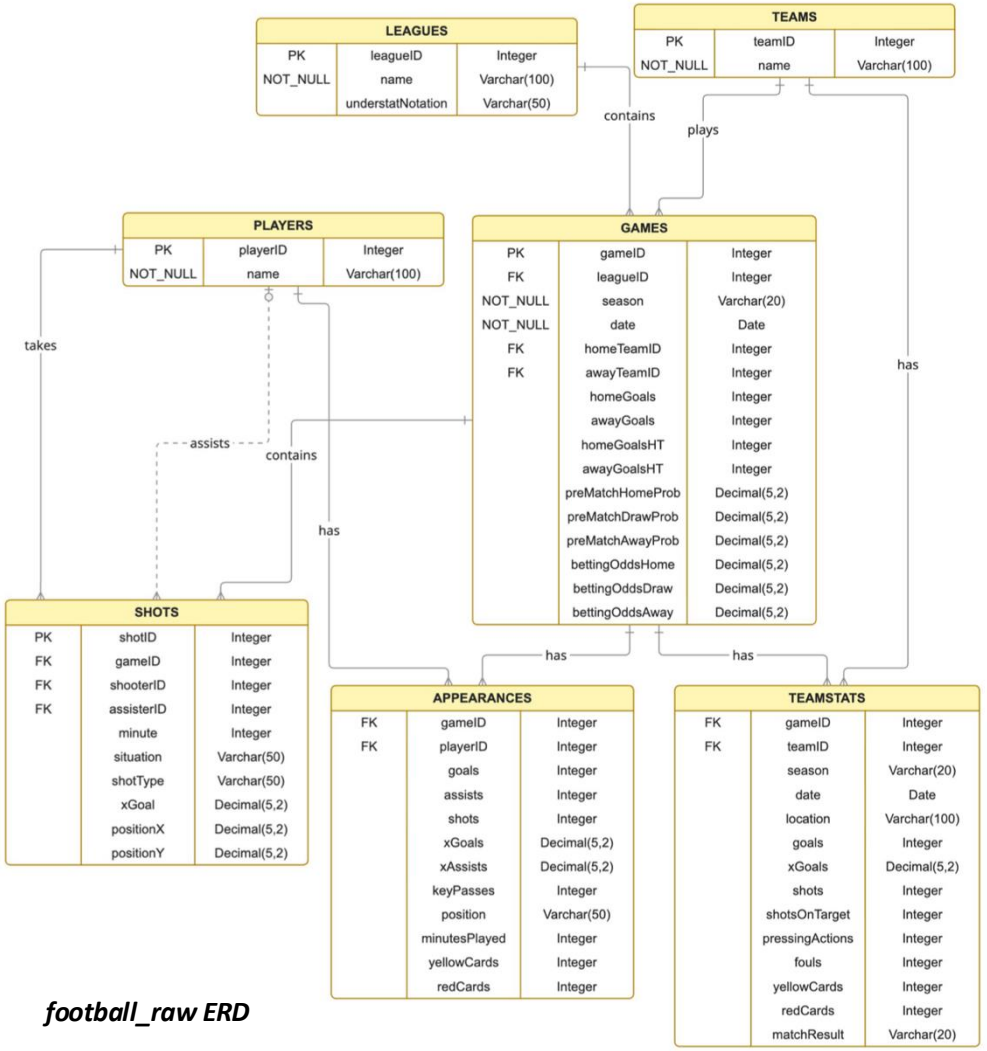
Analytical pain points for brand insights:

1 **No explicit semantic model**

2 **No governance or integrity guarantees**

3 **Redundant & analytically awkward structures**

4 **No standardized analytical layer for brands**



football_raw ERD

A purpose-built relational database converts raw DB into a governed, scalable analytics platform for evidence-based sponsorship decisions

Translate Football reality into structured & trusted data

Encodes real-world structure of football

Real hierarchy: League → Season → Game → Team → Player

Enforces consistency and trust

Relationships & constraints ensure every statistic is tied to a valid entity

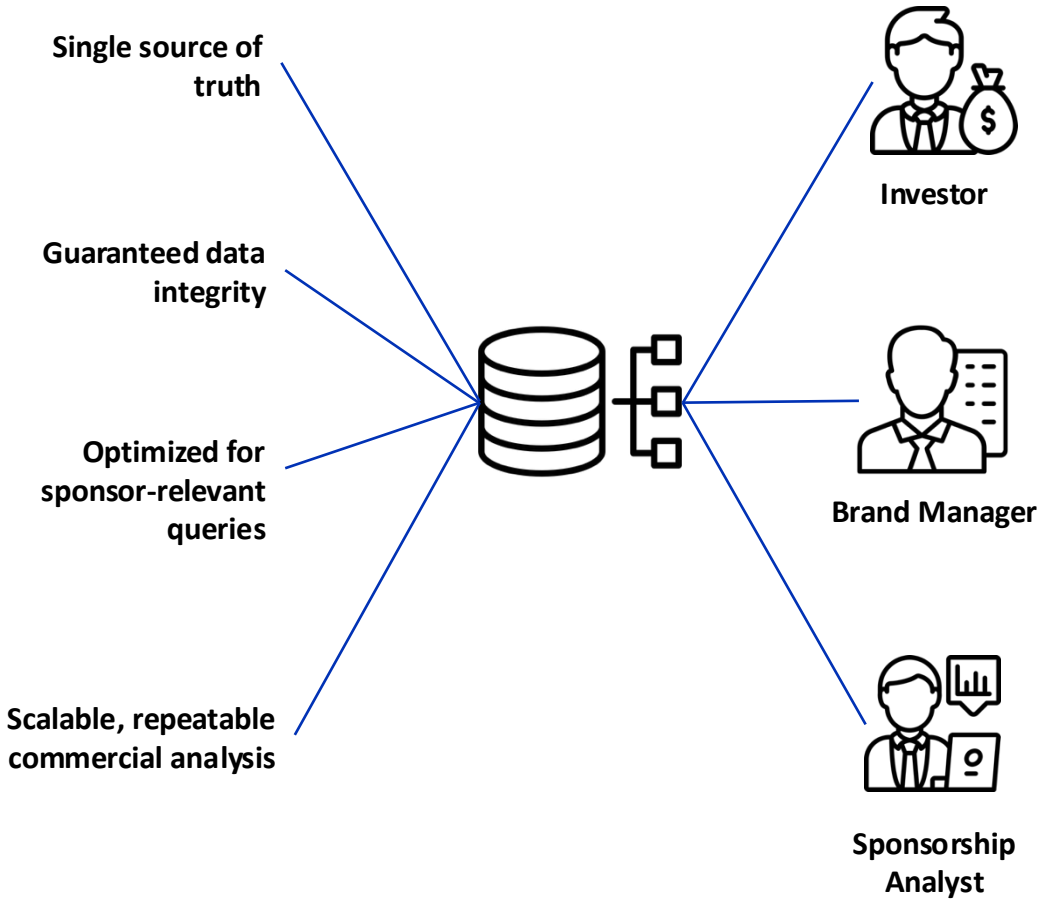
Supports sponsor-relevant analytical patterns

Purpose-built to answer commercial questions

Enables scalable, repeatable analysis

Standardized metrics and reusable queries allow insights to be generated reliably

Unlock commercial confidence in Sponsorship insights



A structured 4-layer pipeline converts raw files into analytics-ready football database

FIFA

1

Raw data ingestion

All CSVs are loaded in original form, without modification

Preserves auditability, sponsors & auditors can trace insights directly back to original source

2

Profiling & validation

Row-counts, key uniqueness, null checks, referential feasibility

Ensures the underlying data is **structurally safe** before being trusted for ROI-linked decisions

3

Core schema structuring

DB normalization & design, data migration, type conversion

Convert raw text into **typed relational tables**, enforce PK-FK, separates game/team/player grains, and restructures betting odds for analytical use

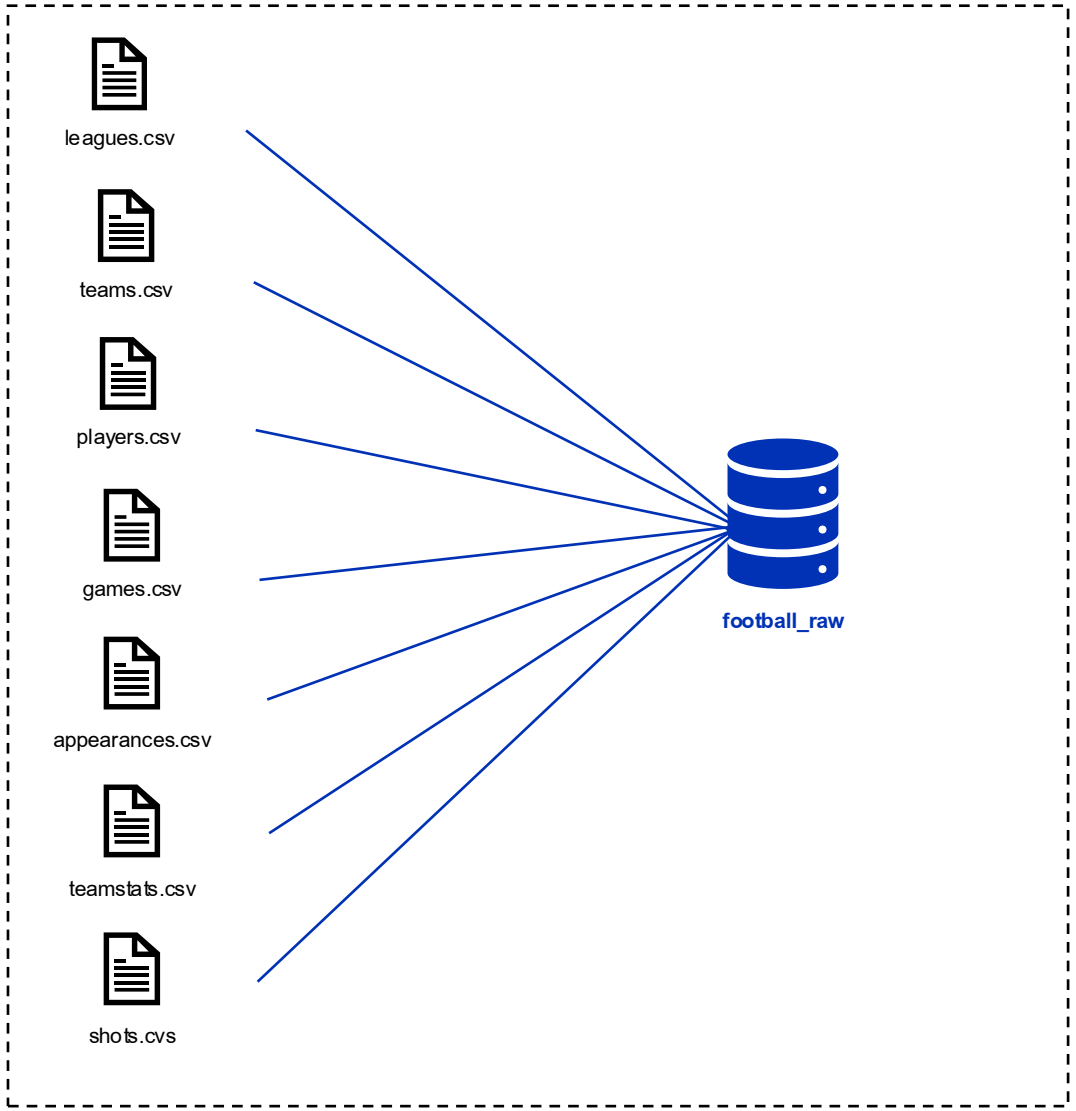
4

Data normalization

Football-specific data normalization for metrics setup

Allows **fair benchmarking of assets across leagues** and seasons so sponsors could compare on equal scales

Step 1: Raw CSVs are ingested into a staging database before transformation



Why the staging layer matters:

Preserve data fidelity

- The raw tables act as an **immutable copy of the source files**.
- No transformations, coercions, or inferred corrections are applied at this stage, allowing the project to maintain a clear audit trail and to re-run transformations if necessary

Isolate parsing errors

- By separating ingestion from transformation, any parsing or type-conversion issues can be detected and handled explicitly rather than silently failing during load.

Support reproducibility

- Analysts can always refer back to football_raw to verify that downstream anomalies (e.g., unexpected NULLs) arise from real gaps in the source rather than transformation artefacts.

Step 2: Profile & Validate the raw layer before designing & normalizing core DB

Row-count reconciliation

Table	Rows
leagues_raw	5
players_raw	7,659
teams_raw	146
games_raw	12,680
appearances_raw	356,513
teamstats_raw	25,360
shots_raw	324,543

=

Primary-key uniqueness confirmed

Distinct IDs
5
7,659
146
12,680
356,513
25,360
324,543

1 For number of rows per table, **all matches the original CSVs**

2 All tables having rows = distinct IDs, proving **IDs are clean and usable as primary keys**

No orphan facts in raw joins

- **0 appearances** without a valid player or game.
- **0 team stats** without a valid team or game.
- **0 shots** without a valid game or shooter.

No blank critical identifiers

- **0 blank/NULL** playerID, teamID, gameID, shooterID, or assisterID in the fact tables checked.

Validation confirms the raw data is complete, coherent, and reliable → providing a safe foundation for building a sponsorship-grade analytical database.

Step 3: Transform the raw data into a normalized analytical schema (football_core)

1 Turning disconnected CSVs into a governed data model:

Convert raw text into typed, governed tables

Raw text fields from raw DB were cast into proper SQL data types

Standardize categorical fields for clean segmentation

Match location (home/away) and results (win/draw/loss)

Enforce referential integrity through relational design

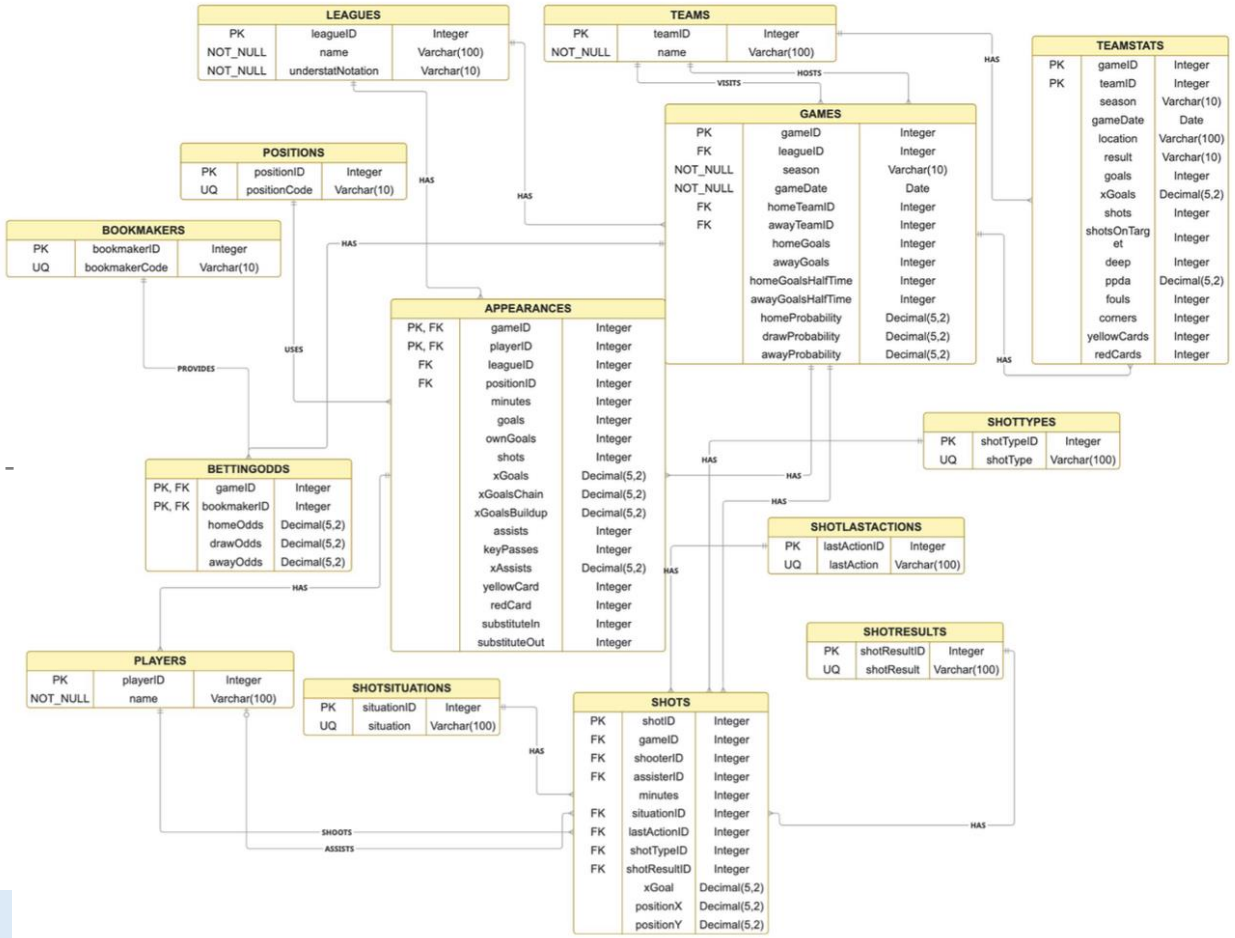
PK & FK constraints were implemented across all core tables

2 Normalizing core schema structure:

Create lookup dimensions for repeated fields

- Positions, shot situations, last actions, shot types, shot results, bookmakers data are stored once, referenced via surrogate IDs
- Restructure bookmaker odds from wide CSV columns (B365H, B365A, etc.) into a vertical table: one row per bookmaker per game

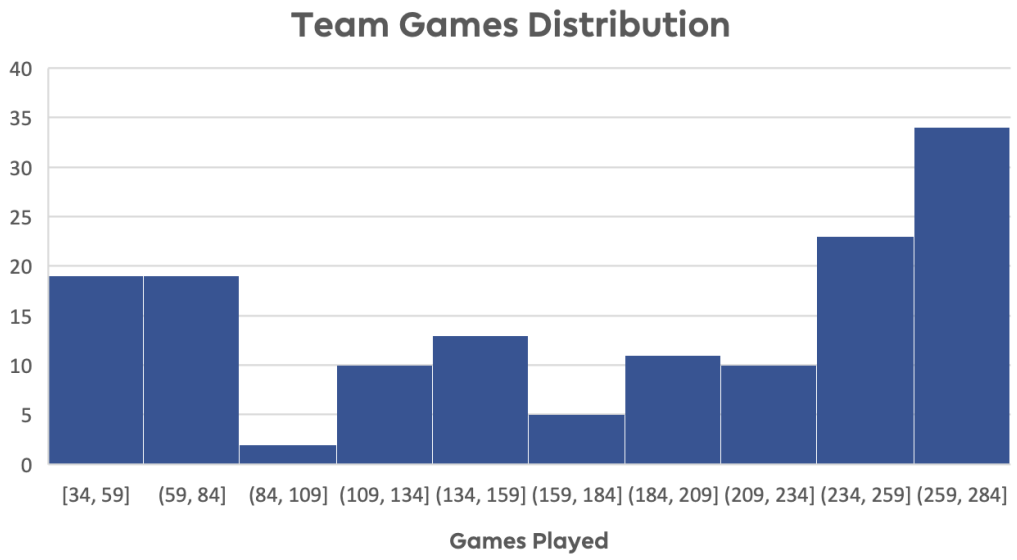
Lighter, consistent, and optimized core DB with 14 entities



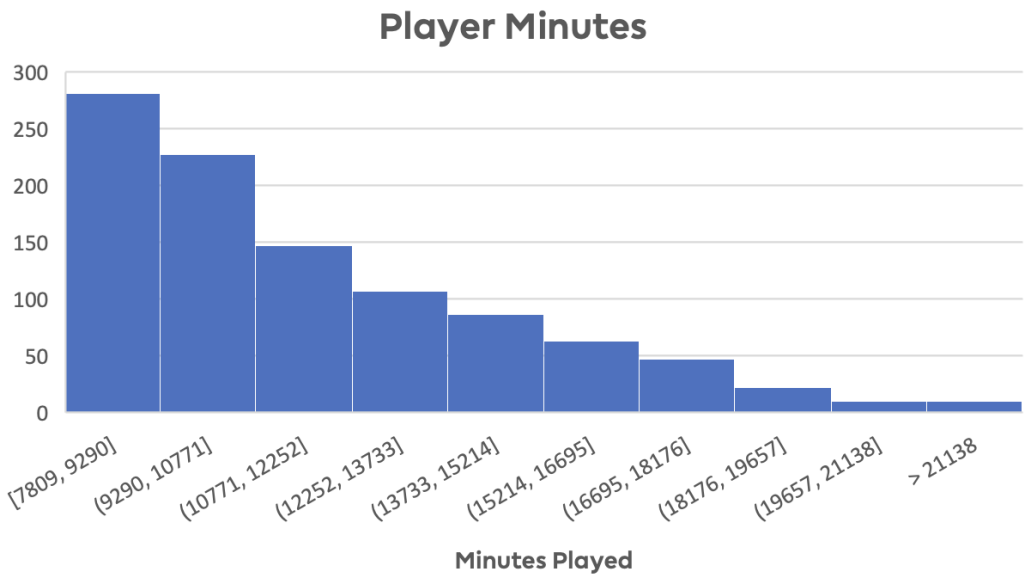
football_core ERD

Step 4: Normalize football stats into fair, comparable sponsor-ready metrics

Teams play different numbers of matches
Raw totals inflate exposure for clubs simply by playing more games



Players accumulate unequal minutes due to rotation
Goals and assists totals mainly reflect time on pitch, not impact



Normalizing metrics **per match** (teams) and **per 90 minutes** (players)

Shifts analysis toward true commercial value - screen time per unit of opportunity - rather than cumulative volume, giving brands a fair way to compare assets before investing.

As results, we have following core entities:

Overall, the database ...

1 Has 14 tables of scalable and robust schema to support:

Descriptive Analytics

enabling reliable aggregation of goals, xG, shots, and participation metrics across players, teams, and leagues.

Tactical and performance analysis

allowing detailed examination of playing styles, chance creation patterns, and player roles through event- and match-level data.

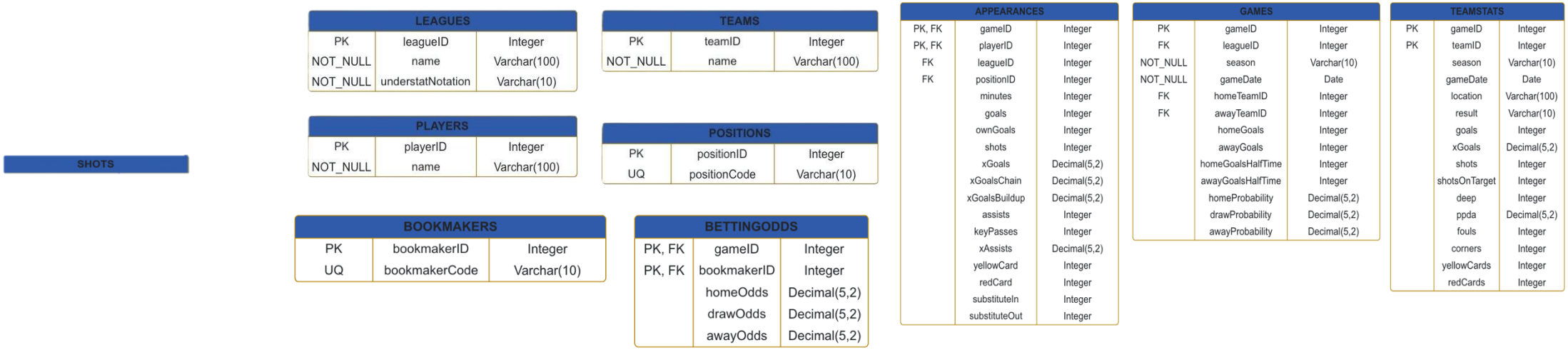
2 Ensures sufficient data volume:

Statistical Power

2 core tables including **Appearances** and **Shots**, contains more than 200 observations on level of player and event analyses.

Cross-analytical through leagues and seasons

supporting analysis of match outcomes and performance trends for business decisions.



As results, we keep following core entities:

Leagues, Teams, Players, Shots, Appearances, Games and **TeamStats**, now **hard-linked through keys** ensuring every KPI is built on verified data, a prerequisite for credible, repeatable sponsorship analytics.

TEAMS

PLAYERS

SHOTS

APPEARANCES		
PK, FK	gameID	Integer
PK, FK	playerID	Integer
FK	leagueID	Integer
FK	positionID	Integer
	minutes	Integer
	goals	Integer
	ownGoals	Integer
	shots	Integer
	xGoals	Decimal(5,2)
	xGoalsChain	Decimal(5,2)
	xGoalsBuildup	Decimal(5,2)
	assists	Integer
	keyPasses	Integer
	xAssists	Decimal(5,2)
	yellowCard	Integer
	redCard	Integer
	substituteIn	Integer
	substituteOut	Integer

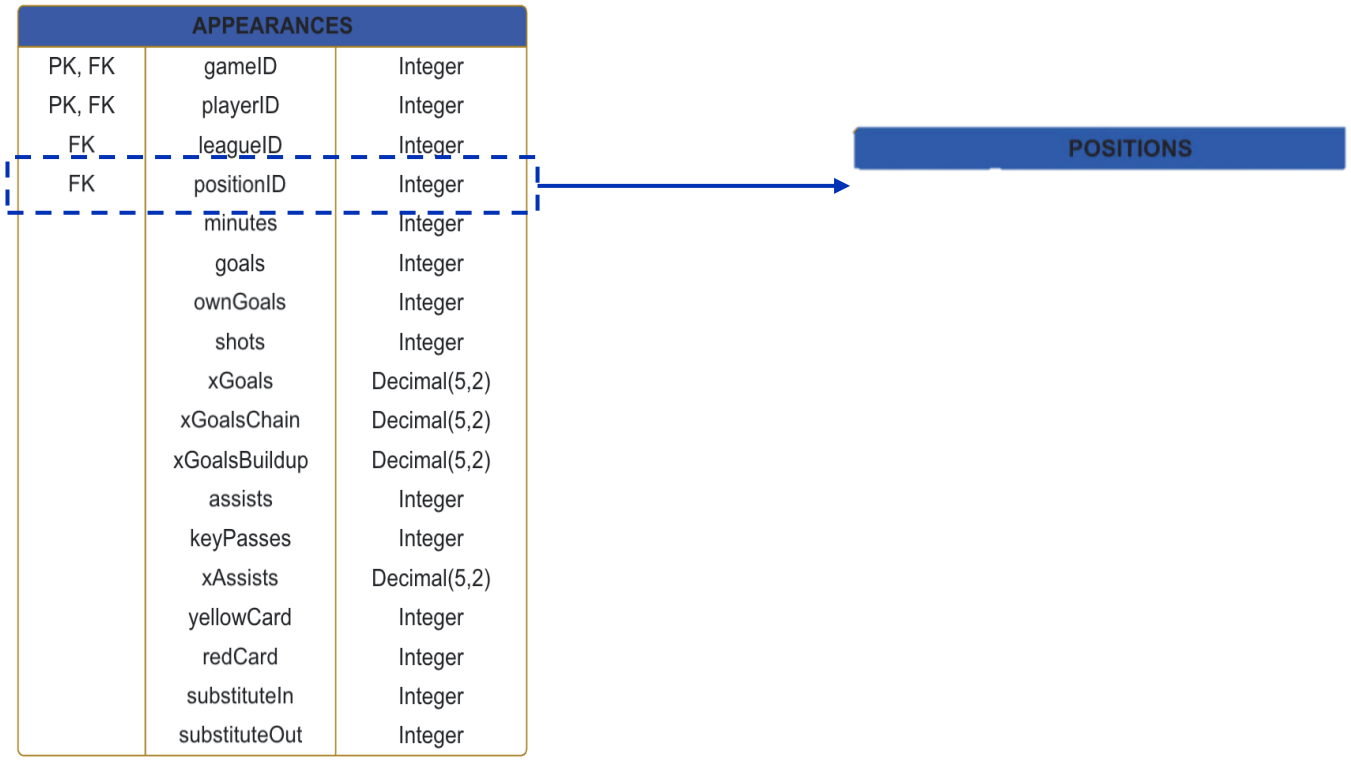
GAMES		
PK	gameID	Integer
FK	leagueID	Integer
NOT_NULL	season	Varchar(10)
NOT_NULL	gameDate	Date
FK	homeTeamID	Integer
FK	awayTeamID	Integer
	homeGoals	Integer
	awayGoals	Integer
	homeGoalsHalfTime	Integer
	awayGoalsHalfTime	Integer
	homeProbability	Decimal(5,2)
	drawProbability	Decimal(5,2)
	awayProbability	Decimal(5,2)

TEAMSTATS		
PK	gameID	Integer
PK	teamID	Integer
	season	Varchar(10)
	gameDate	Date
	location	Varchar(100)
	result	Varchar(10)
	goals	Integer
	xGoals	Decimal(5,2)
	shots	Integer
	shotsOnTarget	Integer
	deep	Integer
	ppda	Decimal(5,2)
	fouls	Integer
	corners	Integer
	yellowCards	Integer
	redCards	Integer

While we normalized the schema with following new look up tables

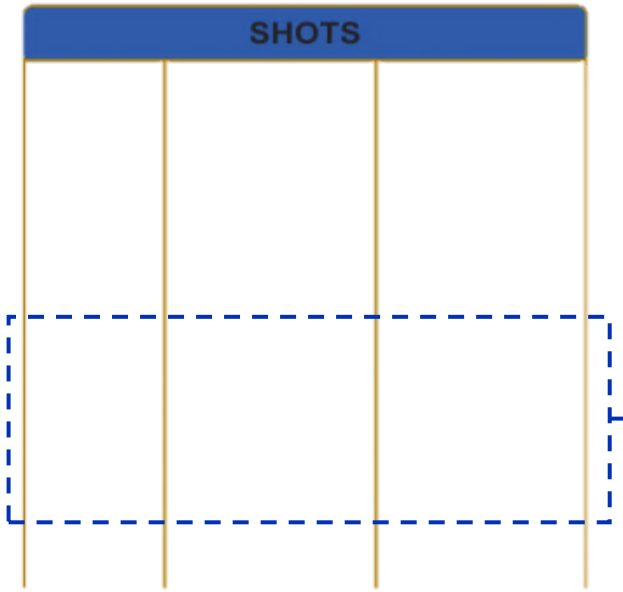
Positions is now a **governed dictionary of player roles**, ensuring every appearance uses a single coded value, enabling clean joins and role-based commercial analysis.

positionID	positionCode
1	GK
2	DR
3	DC
4	DL
5	DMC
6	AMR
7	AMC
8	AML
9	FW
10	Sub
11	MR
12	MC
13	ML
14	FWR
15	FWL
16	DMR
17	DML



While we normalized the schema with following new look up tables

Shot events are decomposed into **four lookup entities: ShotSituations, ShotLastActions, ShotTypes, ShotResults,** each mapped via IDs inside the Shots fact table, transforming messy text fields into analyzable commercial data



lastActionID	lastAction
1	Standard
2	Pass
3	Tackle
4	BallRecovery
5	None
6	Cross
7	Chipped
8	Rebound
9	Aerial
10	Dispossessed
11	Throughball

⋮

situationID	situation
1	DirectFreekick
2	SetPiece
3	OpenPlay
4	FromCorner
5	Penalty

SHOTSITUATIONS

shotResultID	shotResult
1	BlockedShot
5	Goal
2	MissedShots
4	OwnGoal
3	SavedShot
6	ShotOnPost

shotTypeID	shotType
1	LeftFoot
2	RightFoot
3	Head
4	OtherBodyPart

While we normalized the schema with following new look up tables

games	
gameID	integer
leagueID	integer
season	varchar(10) NN
gameDate	date NN
homeTeamID	integer
awayTeamID	integer
homeGoals	integer NN
awayGoals	integer NN
homeGoalsHalfTime	integer
awayGoalsHalfTime	integer
homeProbability	decimal(6,5)
drawProbability	decimal(6,5)
awayProbability	decimal(6,5)
B365H	decimal(6,2)
B365D	decimal(6,2)
B365A	decimal(6,2)
BWH	decimal(6,2)
BWD	decimal(6,2)
BWA	decimal(6,2)
IWH	decimal(6,2)
IWD	decimal(6,2)
IWA	decimal(6,2)
PSH	decimal(6,2)
PSD	decimal(6,2)
PSA	decimal(6,2)
WHH	decimal(6,2)
WHD	decimal(6,2)
WHA	decimal(6,2)
VCH	decimal(6,2)
VCD	decimal(6,2)
VCA	decimal(6,2)
PSCH	decimal(6,2)
PSCD	decimal(6,2)
PSCA	decimal(6,2)

Every bookmaker & its odds appeared as its own set of wide columns (B365H, B365D, B365A, WHH, WHD, WHA, etc.), duplicating structure across 75,000+ rows and making odds analysis nearly impossible.

The relational model restructures all odds into **a vertical fact table** (BettingOdds) keyed by (gameID, bookmakerID), supported by a **Bookmakers lookup table.**

BOOKMAKERS		
PK	bookmakerID	Integer
UQ	bookmakerCode	Varchar(10)

BETTINGODDS		
PK, FK	gameID	Integer
PK, FK	bookmakerID	Integer
	homeOdds	Decimal(5,2)
	drawOdds	Decimal(5,2)
	awayOdds	Decimal(5,2)

GAMES		
PK	gameID	Integer
FK	leagueID	Integer
NOT_NULL	season	Varchar(10)
NOT_NULL	gameDate	Date
FK	homeTeamID	Integer
FK	awayTeamID	Integer
	homeGoals	Integer
	awayGoals	Integer
	homeGoalsHalfTime	Integer
	awayGoalsHalfTime	Integer
	homeProbability	Decimal(5,2)
	drawProbability	Decimal(5,2)
	awayProbability	Decimal(5,2)

The Five Pillars

1 **Audience Exposure:** Quantifying reach via attacking actions (shots, goals) as proxies for screen time.

2 **Narrative Impact:** Leveraging seasonal predictability trends to align with high-resonance fan moments.

3 **Reliability:** Assessing year-to-year and venue-to-venue consistency for predictable ROI.

4 **Risk Management:** Correlating impact with disciplinary risk and playing-time volatility.

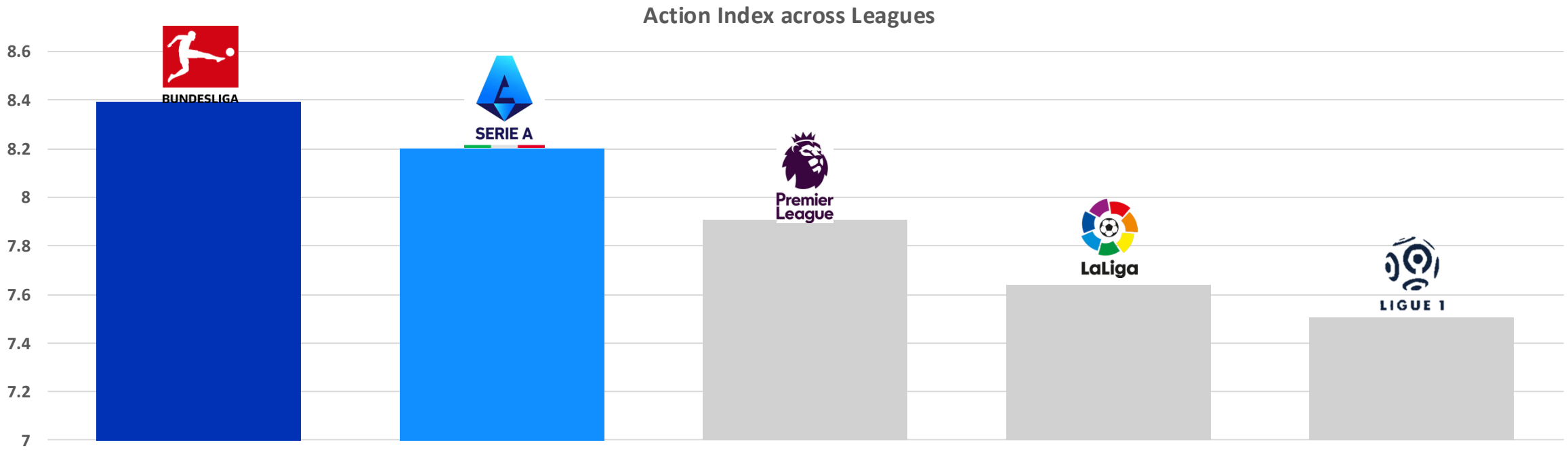
5 **Value Identification:** Using xG and xA to detect "Hidden Stars" before market valuation peaks.

Question 1: On-pitch action exposure & Brand Visibility efficiency

Comparative action intensity across Europe's top leagues, measuring expected sponsor logo exposure per match through goals, shots, and xG.

Business Question: *How do leagues rank by on-pitch action exposure per match?*
Commercial Rationale: *Identifying leagues where each match naturally generates more high-attention attacking moments, maximizing repeated logo exposure per game across broadcasts, highlights, and digital clips regardless of audience scale.*

Action intensity,
declines steadily from the Bundesliga to Ligue 1



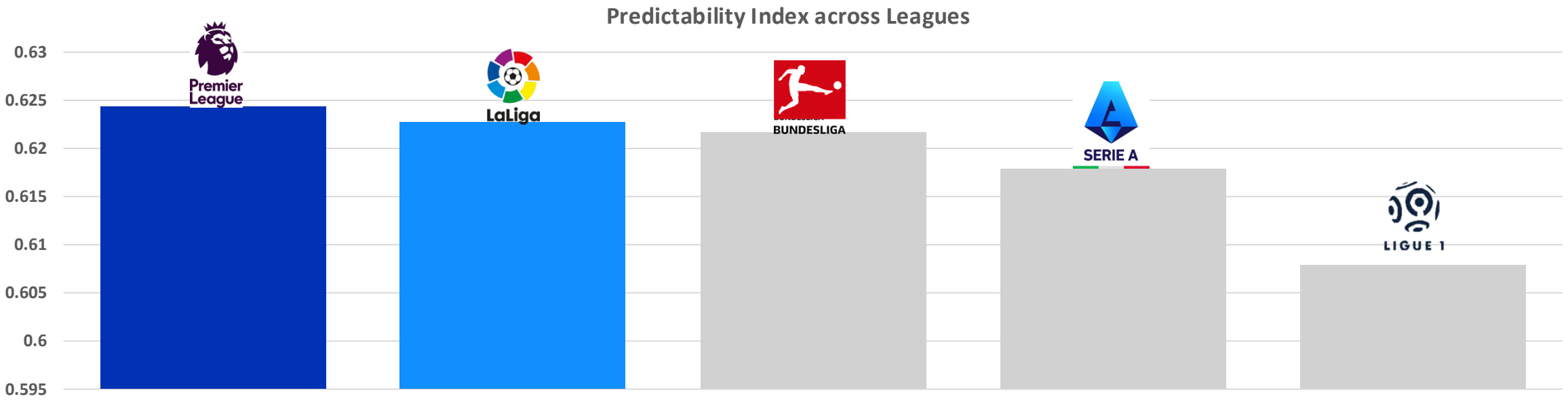
Prioritize high-action leagues for visibility efficiency: Determine which league's sponsored match delivers more frequent attacking moments and repeated logo exposure, making them ideal for sponsors seeking **maximum visibility per game**.
Balance exposure efficiency with scale in portfolio planning: While the Premier League offers slightly lower action intensity, its strong position complement high-action leagues to combine efficient per-match exposure with global reach and commercial depth.

Question 2: Match outcome predictability across Leagues

Predictability Index shows whether matches tend to follow expectations or produce frequent upsets that shape fan engagement.

Business Question: *How do leagues rank by match outcome predictability?*
Commercial Rationale: Decide whether to anchor campaigns in leagues with reliable, expectation-confirming narratives or leverage more volatile leagues where surprise results create emotionally charged moments and viral storytelling opportunities.

All leagues show,
similar predictability levels with only small differences between them.



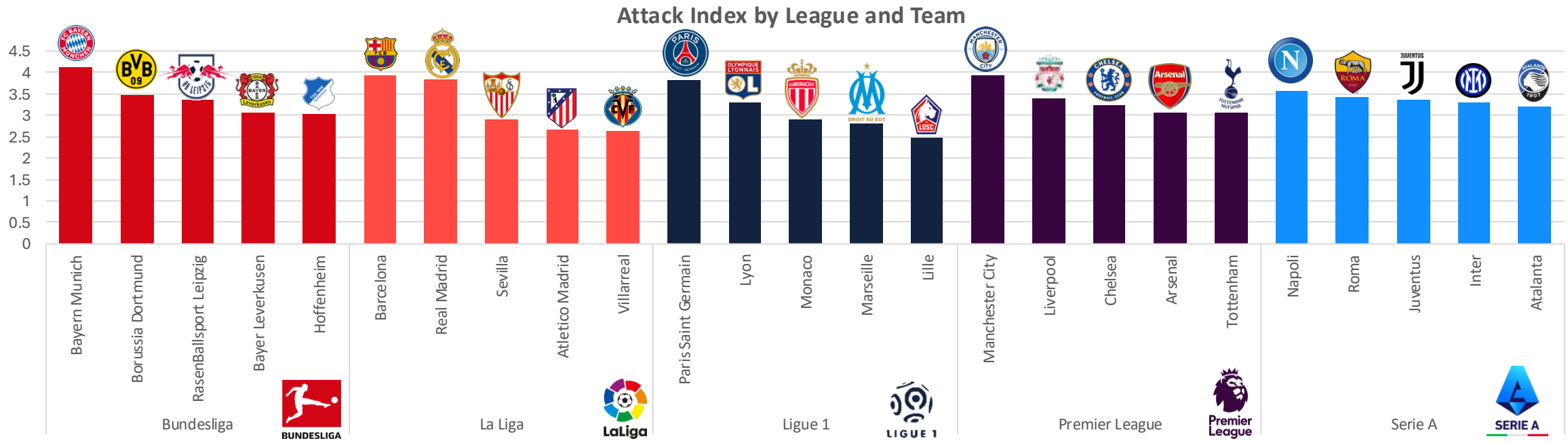
Use high-predictability leagues for narrative stability: The **Premier League** and **La Liga** rank highest in predictability, making them well-suited for long-term brand storytelling, consistent messaging, and expectation-aligned sponsorship campaigns.
Leverage low-predictability leagues for emotional impact: Ligue 1 and **Serie A** show lower predictability, offering greater upset potential that brands can exploit for high-emotion, viral moments and short-burst, attention-driven activations.

Question 3: Top 5 teams ranked by attacking volume within each league

Attacking volume predicts how often a club appears in goals, chances, and highlight packages, maximizing sponsor logo exposure across digital replays & broadcasts

Business Question: *Who are the top 5 teams ranked by attacking volume within each league?.*
Commercial Rationale: *Identifying clubs most likely to feature in broadcast highlights and post-match media due to their consistently high volume of attacking actions, making them prime candidates for high-visibility sponsorship placements.*

Across all leagues,
a small group of top teams consistently dominate the Attack Index, highlighting the highly concentrated



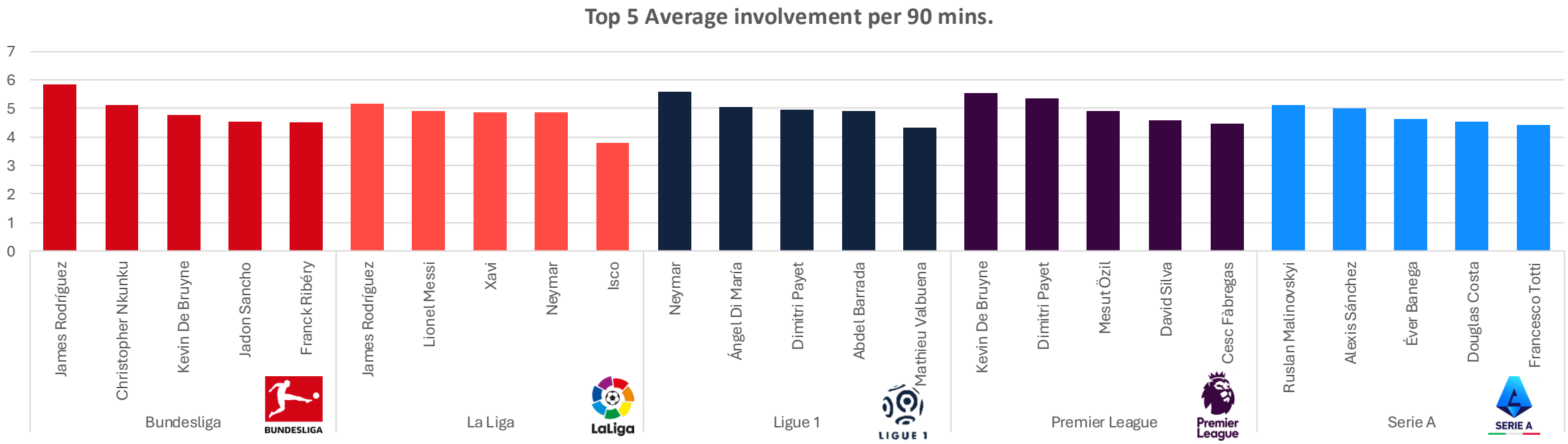
Dynamic Strategy: Activate sponsorships around high-attack teams during marquee fixtures to capitalize on intense, highlight-rich matches that drive immediate attention.
Brand Alignment: Pair brands seeking excitement and aggression with high Attack Index teams, while positioning more heritage or control-oriented brands alongside mid-ranked teams for steadier, less volatile exposure.

Question 4: Top 5 players ranked by on-pitch involvement per 90 minutes

Captures continuous on-ball involvement that directly drives broadcast visibility and repeated brand exposure beyond isolated scoring moments.

Business Question: *Who are the top 5 players ranked by on-pitch involvement per 90 minutes?.*
Commercial Rationale: *Identify players who maximize continuous on-camera presence and brand exposure by staying deeply involved in attacking play, rather than merely scoring goals.*

Most commercially valuable players,
are not always the top scorers, but creative midfielders and playmakers who consistently drive possession and attacking sequences.



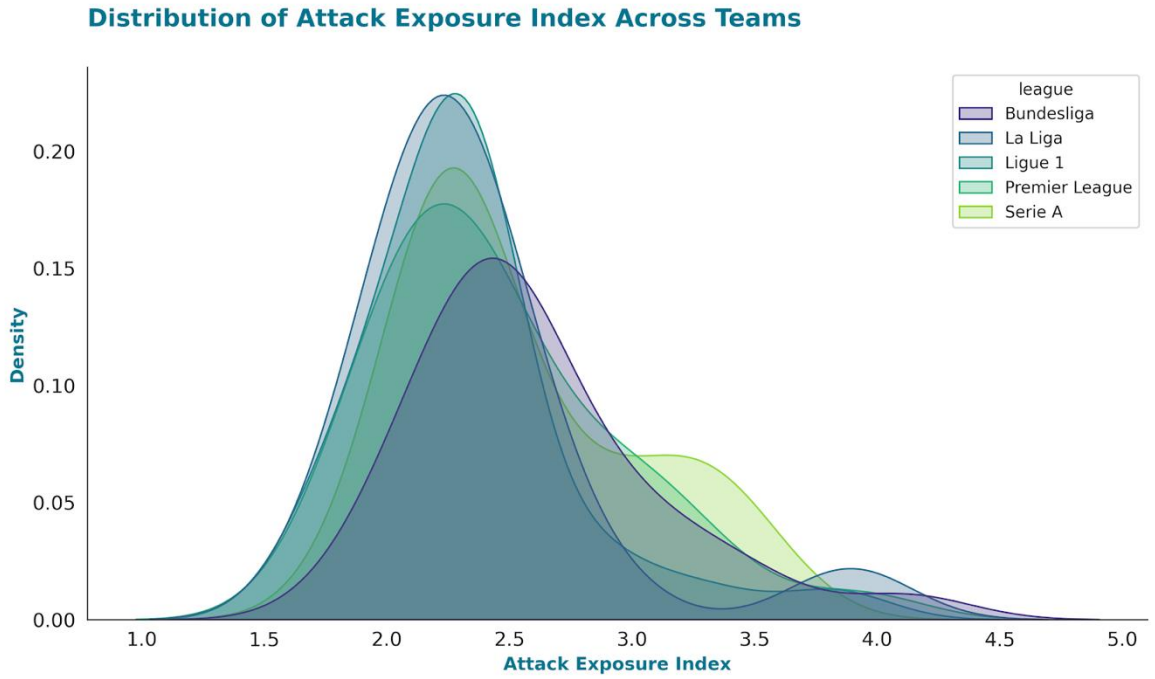
Activation strategy: Design in-match activations (replays, lower-third graphics, social cut-ins) around high-involvement players, as they appear repeatedly in buildup and chance-creation phases rather than only at scoring moments.
Risk-hedging strategy: Prioritize players with high involvement per 90 *and* sustained minutes played to reduce exposure volatility from injuries, substitutions, or form drops while maintaining consistent broadcast presence.

Question 5: Distribution of Attacking Exposure Across Teams

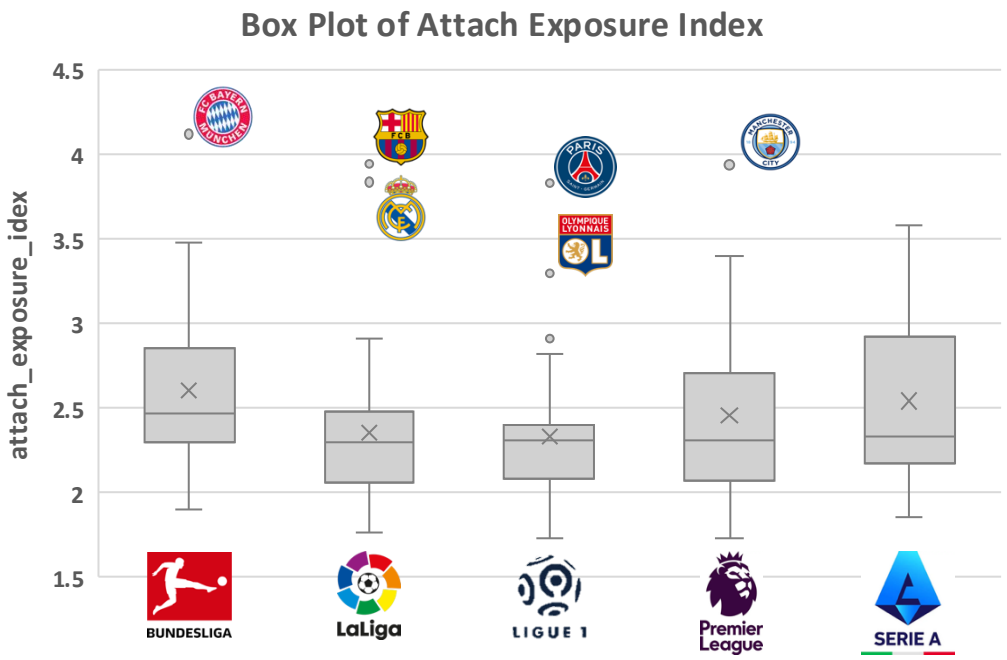
Attacking visibility is moderately concentrated, allowing brands to choose between high-tier exclusivity with "Elite Outliers" or risk-diversified mid-tier portfolios

Business Question: *How is attacking visibility distributed across clubs within and across the Top-5 European leagues?.*
Commercial Rationale: Identifying whether fan attention is concentrated on a small elite or evenly spread dictates whether brands should pursue exdusive single-club deals or multi-team reach campaigns.

Most clubs sit in a core cluster,
with a smaller group extending into a high-intensity tail.



Elite outliers (e.g., Bayern Munich, PSG)
provide visibility far exceeding the league median.



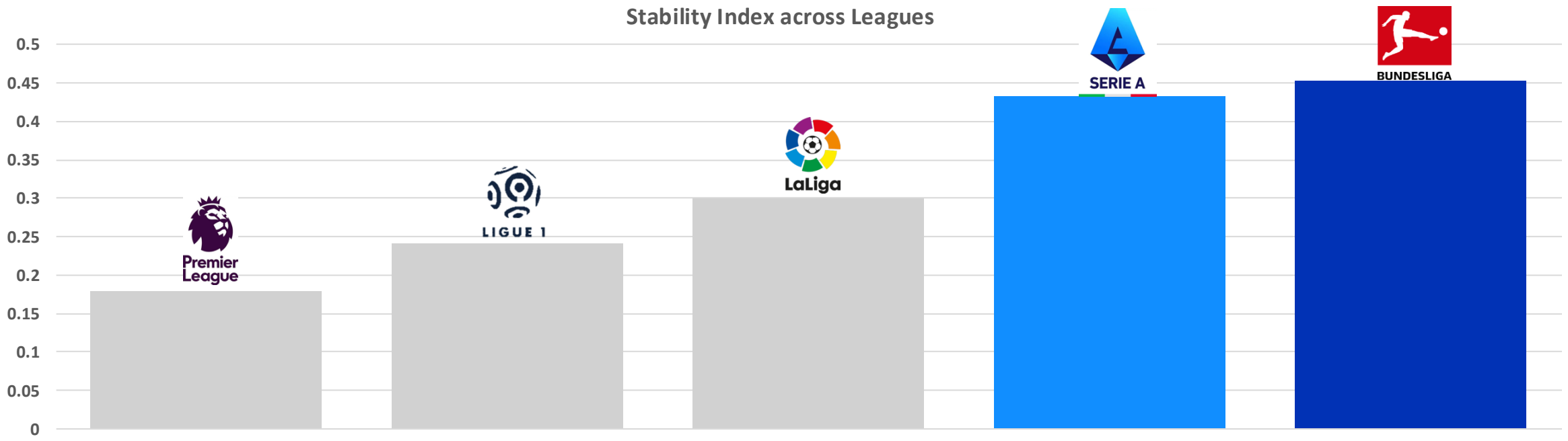
Premium Strategy: Seek exclusive deals with high-exposure elite clubs to maximize broadcast screen time and perceived dominance.
Portfolio Strategy: Bundle several mid-tier clubs to approximate the aggregate reach of an elite club while diversifying performance and market risk.

Question 6: Leagues rank by stability

It reveals whether sponsorship exposure can be reliably forecast over multiple seasons or is subject to large, unpredictable swings that increase investment risk.

Business Question: *How do leagues rank by year-to-year stability?.*
Commercial Rationale: *Decide whether to commit to long-term sponsorships in stable leagues with predictable exposure or adopt flexible, short-term deals in more volatile leagues where returns fluctuate year to year.*

Premier League shows the highest multi-season stability in attacking output,
Serie A and the Bundesliga exhibit much greater year-to-year volatility.



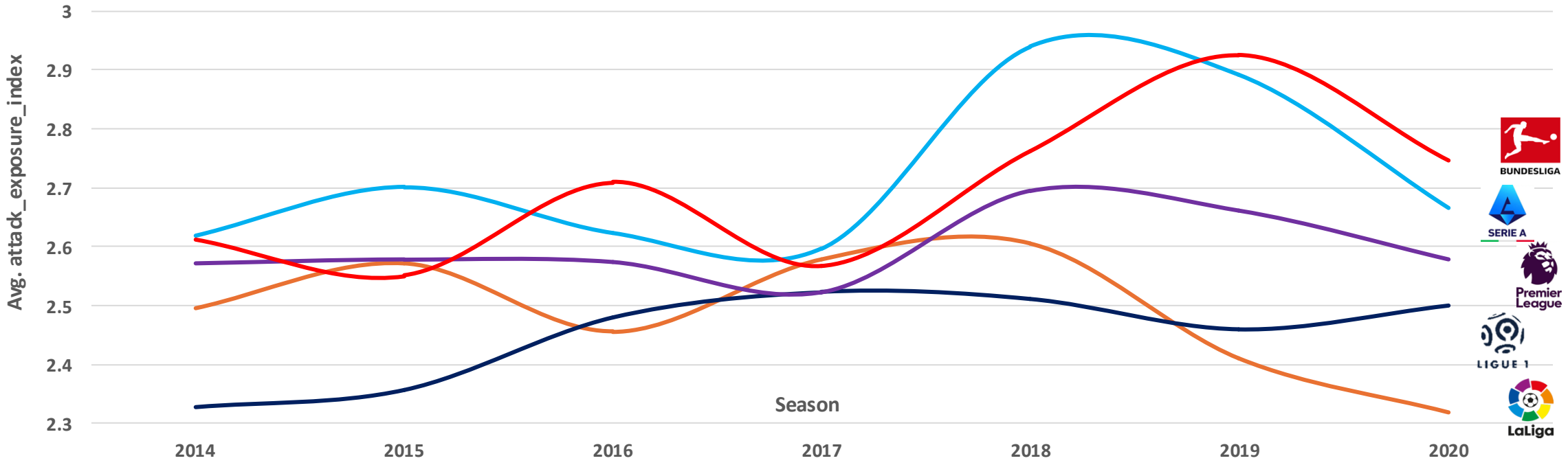
Premium strategy: Commit to long-term, multi-year sponsorships in highly stable leagues like the Premier League to confidently project consistent exposure and ROI over time.
Flexible strategy: In more volatile leagues such as Serie A and the Bundesliga, favor shorter-term or annually adjustable deals to manage exposure risk and adapt to shifting league dynamics.

Question 7: Stability of Team Attacking Exposure Over Time

Long-term sponsorship ROI depends on the durability of a club's attacking identity rather than short-term performance spikes.

Business Question: *Is a club's attacking visibility a durable attribute or a product of short-term spikes?*
Commercial Rationale: Brands planning multi-season partnerships need confidence that a club's exposure will not collapse after one strong year. Identifying repeatable attacking presence helps isolate dependable long-term assets.

Significant year-to-year volatility across all five leagues,
Bundesliga and Serie A display higher attack exposure levels over time.



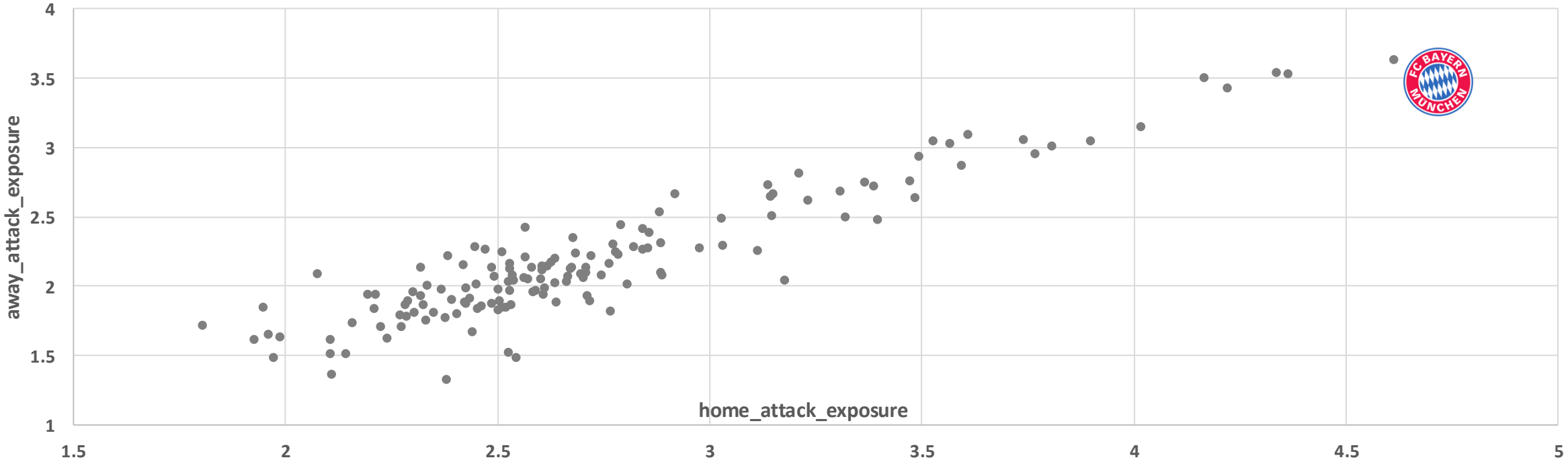
Asset Selection: Target clubs with repeatable, season-to-season attacking presence to secure dependable long-term brand exposure.
Risk Management: Use shorter, performance-linked, or flexible sponsorship structures for highly volatile teams.
Contractual Structuring: Stable assets justify multi-year commitments, whereas volatile assets require protective clauses against exposure swings.

Question 8: Home vs. Away Attacking Output Across Teams

Teams with venue-independent attacking identities offer more consistent on-screen exposure and higher sponsorship reliability.

Business Question: *How does attacking visibility differ between a team's home and away fixtures?*
Commercial Rationale: Identifying the home-away split determines whether a brand should prioritize stadium-centric assets (if home-driven) or broad

Clear home advantage exists across all leagues,
but the strong positive correlation indicates that elite attacking identities generally remain active regardless of venue.



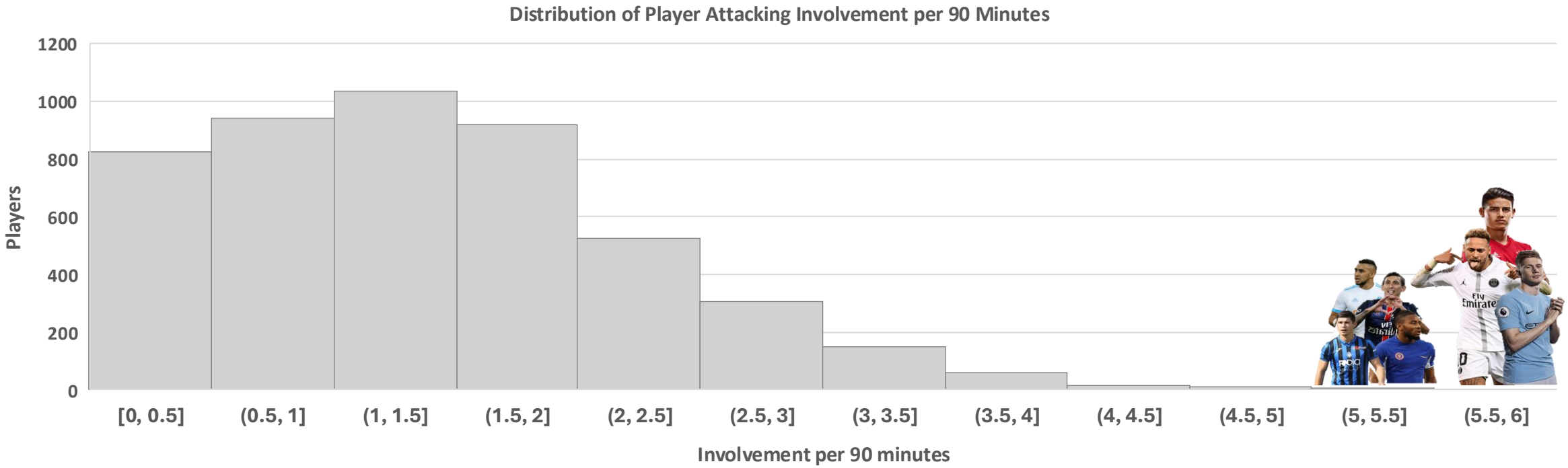
Prioritize "Travelers": Invest in teams with venue-independent attacking output to ensure brand logos appear consistently in both home and away broadcast feeds.
Asset Tailoring: Utilize stadium-centric assets (LED boards, local activations) for teams with heavy home-bias exposure.
Broadcast Focus: Target teams with high away exposure to maximize global reach and impressions through neutral and away-market broadcasts.

Question 9: Distribution of Player Involvement within Leagues

A right-skewed distribution of player involvement reveals a small elite of "High-Impact Creators" who offer premium, concentrated brand exposure.

Business Question: *How evenly is on-pitch involvement shared across players within leagues?*
Commercial Rationale: Identifying whether a league has a broad pool of high-visibility individuals or a small dominant elite dictates ambassador strategy. Concentrated involvement drives scarcity and higher fees for a selective few, while broad distribution supports lower-cost influencer portfolios.

The right-skewed distribution
highlights that high-impact creators like James Rodriguez, Neymar, and Kevin De Bruyne are rare, premium brand assets.



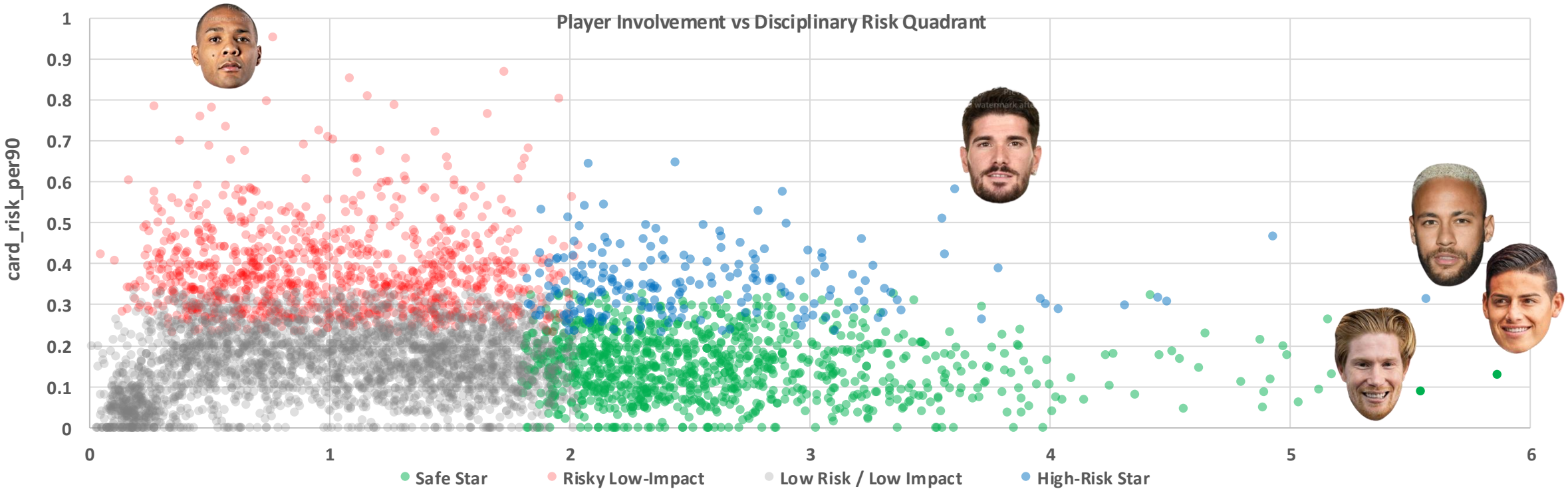
Premium Focus: Prioritize "Long Tail" stars to maximize screen-time-per-minute.
Data-First Selection: Use per-90 metrics to bypass reputational bias and identify high-frequency broadcast actors.
Portfolio Scaling: Bundle mid-tier creators in broad-distribution leagues for equivalent reach at a lower cost-per-impression.

Question 10: Player Involvement vs. Disciplinary Risk

Optimal sponsorship assets—"Safe Stars"—combine high on-pitch involvement with controlled discipline to minimize reputational and availability risk.

- **Business Question**: *How does disciplinary risk correlate with player involvement?*
- **Commercial Rationale:** Identifying behavioral risk alongside performance metrics safeguards campaign continuity and brand sentiment.

Only a specialized subset of high-impact players maintains the controlled discipline required for stable, long-term brand ambassadorship.



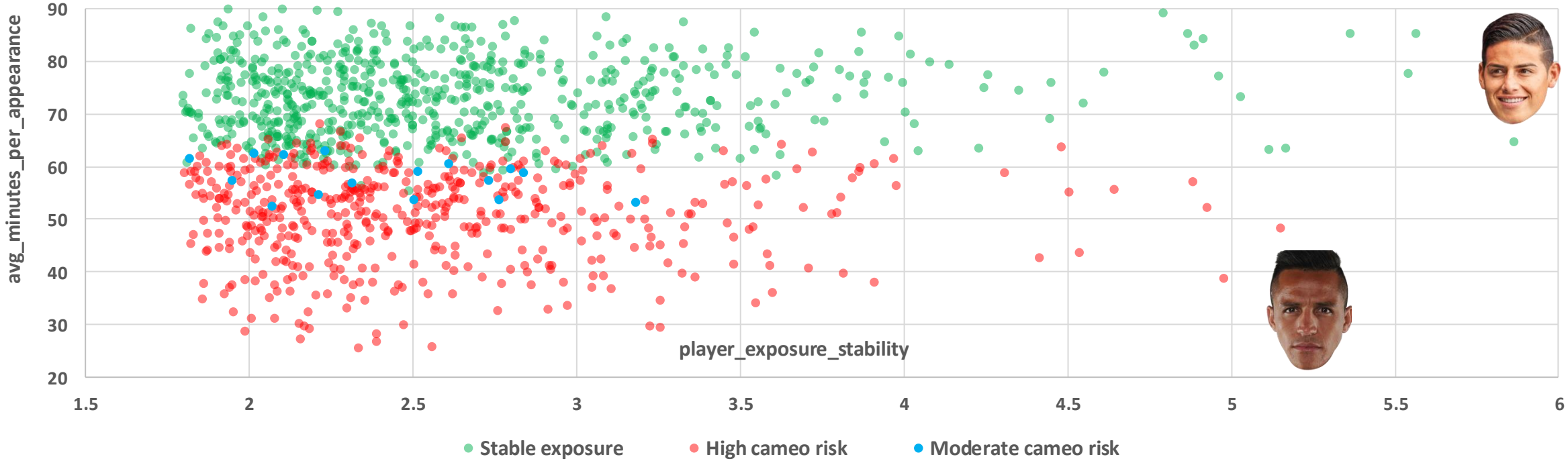
- **Prioritize "Safe Stars"**: Focus investment on elite assets like **James Rodríguez** and **Kevin De Bruyne** who combine high impact with controlled discipline.
- **Risk-Adjust Valuation**: Apply a commercial discount to "High-Risk Stars" like **Neymar** to account for suspension-driven inactivity.
- **Safeguard Integrity**: Proactively screen for behavioral risk to protect campaign continuity and brand reputation.

Question 11: Player Exposure Reliability vs. Playing Time

Analytical impact must be paired with playing time reliability to avoid the "Sponsorship Trap" of high-talent, low-visibility assets.

Business Question: *How often do high-involvement players fail to reach meaningful screen time?*
Commercial Rationale: Identifying exposure volatility helps brands avoid over-indexing on players whose statistical quality does not translate into consistent camera time.

A significant cluster of elite creators shows high per-90 involvement but low average minutes, indicating high "Cameo Risk" that undermines sponsorship ROI.



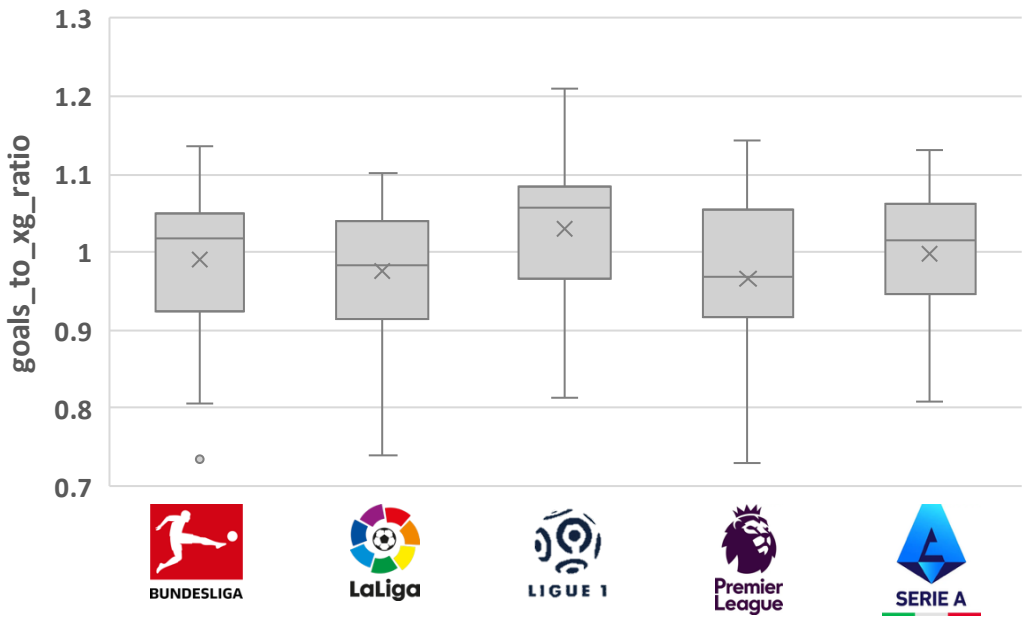
- **Prioritize Stability**: Invest in high-involvement stars with high average minutes per match to ensure consistent logo visibility.
- **Discount "Cameo" Assets**: Lower valuations for rotation players with strong per-90 metrics but unreliable screen presence.
- **Validate Playing Time**: Use average-minutes-per-match data to distinguish between dependable starters and high-impact bench options.

Question 12: Team Expected vs. Realized Outcomes

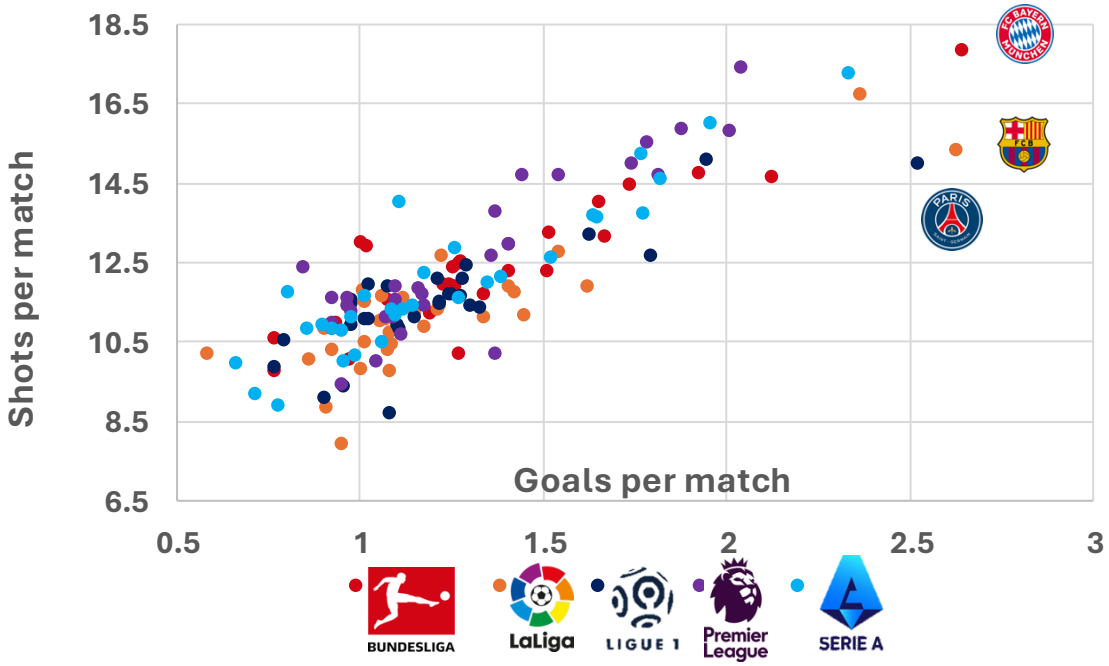
Identifying the gap between expected goal and actual goals uncovers undervalued "hidden stars" and unsustainable "hype" risks.

Business Question: *How does expected attacking output compare to realized outcomes across teams?*
Commercial Rationale: *xG* represents forward-looking quality; teams under-performing their *xG* are "hidden" value opportunities, while over-performers face regression risk.

Most leagues cluster around a 1.0 ratio,
but a long tail of under-estimated reveals untapped value.



Higher shot volume generally
translates to more goals, reflecting sustained pressure.



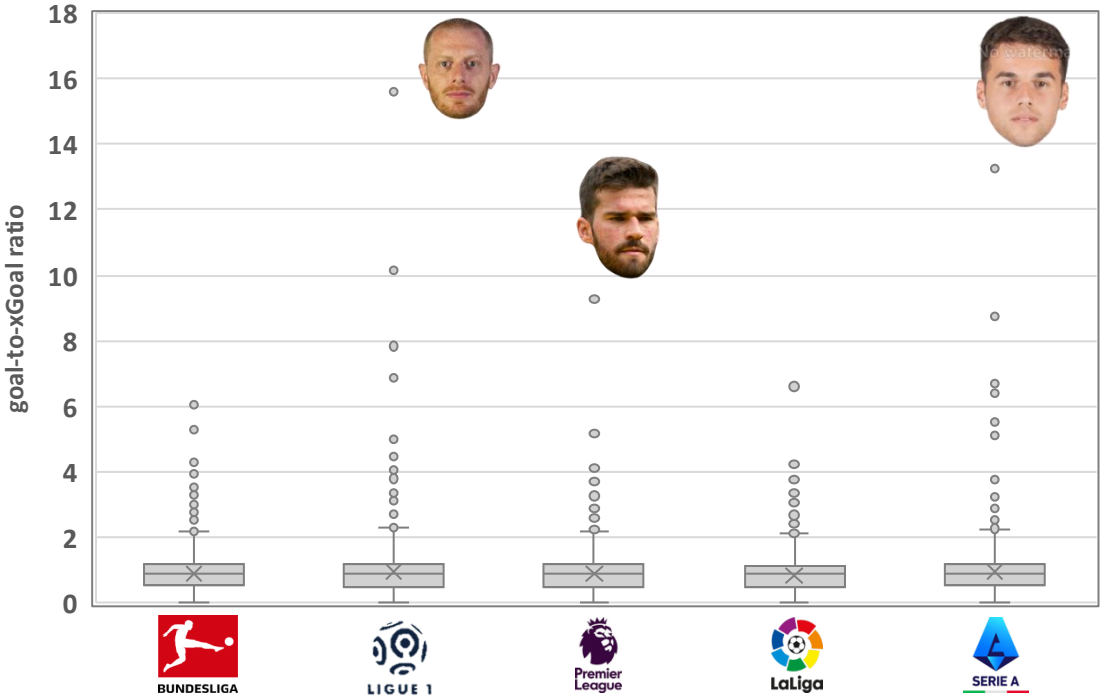
- **Target "Hidden Stars"**: Invest in xG under-performers for high-upside partnerships before market prices rise.
- **Validate Premiums**: Only pay top-tier fees for teams that consistently convert attacking volume into realized goals.
- **Manage Regression**: Exercise caution with teams scoring far above expectation, as unsustainable results may lead to future exposure declines.

Question 13: Player Expected vs. Realized Outcomes

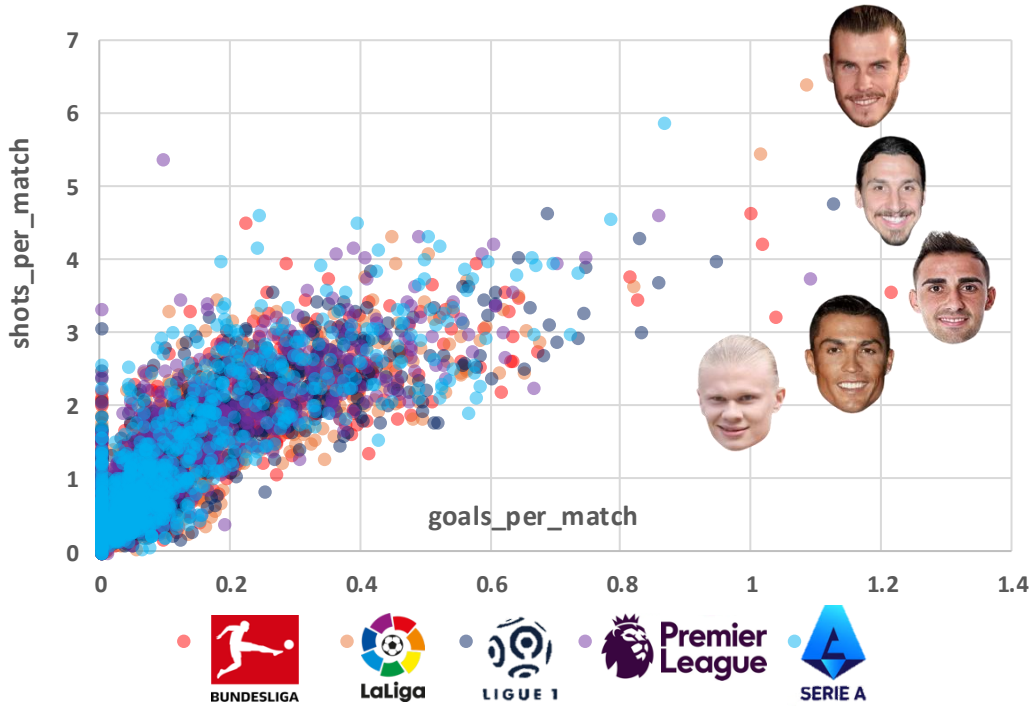
Identifying the gap between underlying attacking quality and actual goals uncovers undervalued assets and unsustainable "hype" risks.

Business Question: *How does expected attacking output compare to realized outcomes across players?*
Commercial Rationale: *xG* represents forward-looking quality; players under-performing their *xG* are "hidden" value opportunities, while over-performers face regression risk.

Over-performance is rare and tightly clustered,
while a "long tail" of under-estimated suggests significant untapped value .



Strong xG correlation confirms talent quality,
but dispersion reveals "Hidden Stars"



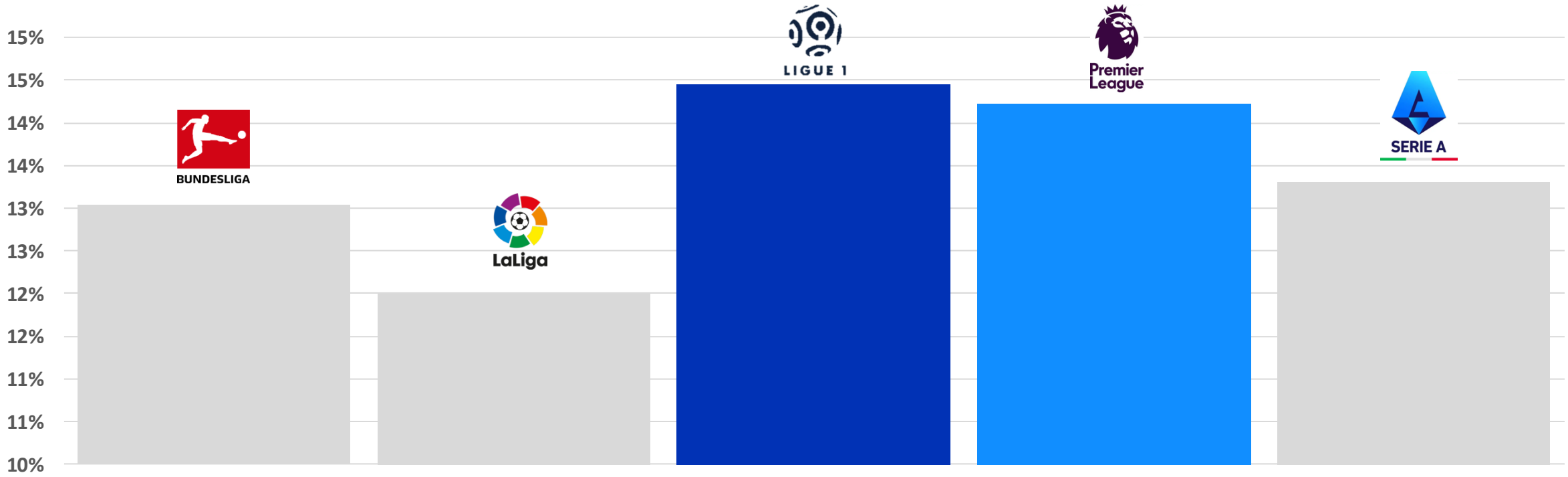
- **Target "Hidden Stars":** Contract players with high goal_to_xG ratio to capture value before their market perception catches up to their quality.
- **Devalue Over-performance:** Exercise caution with "hype assets" scoring significantly above xG, as their commercial exposure may be short-lived.
- **Invest in Untapped Value:** Utilize the "long tail" of under-performance to build a high-potential ambassador portfolio at a fraction of the cost of established elites.

Question 14: Competitive Volatility by League

Align sponsorship strategy to volatility: emotional leagues drive engagement, stable leagues deliver dependable reach.

Business Question: How do leagues differ in the frequency with which underdogs beat betting market expectations?
Commercial Rationale: Upset frequency determines emotional tone: high-volatility leagues fuel shock-win storytelling, while stable leagues support consistent, long-term brand messaging. Knowing this lets sponsors time spend strategically, chase emotion or bank on predictability.

Frequency of underdogs beating the market expectations,
Ligue 1 and Premier league leads the ranking in terms of “magic” match outcomes



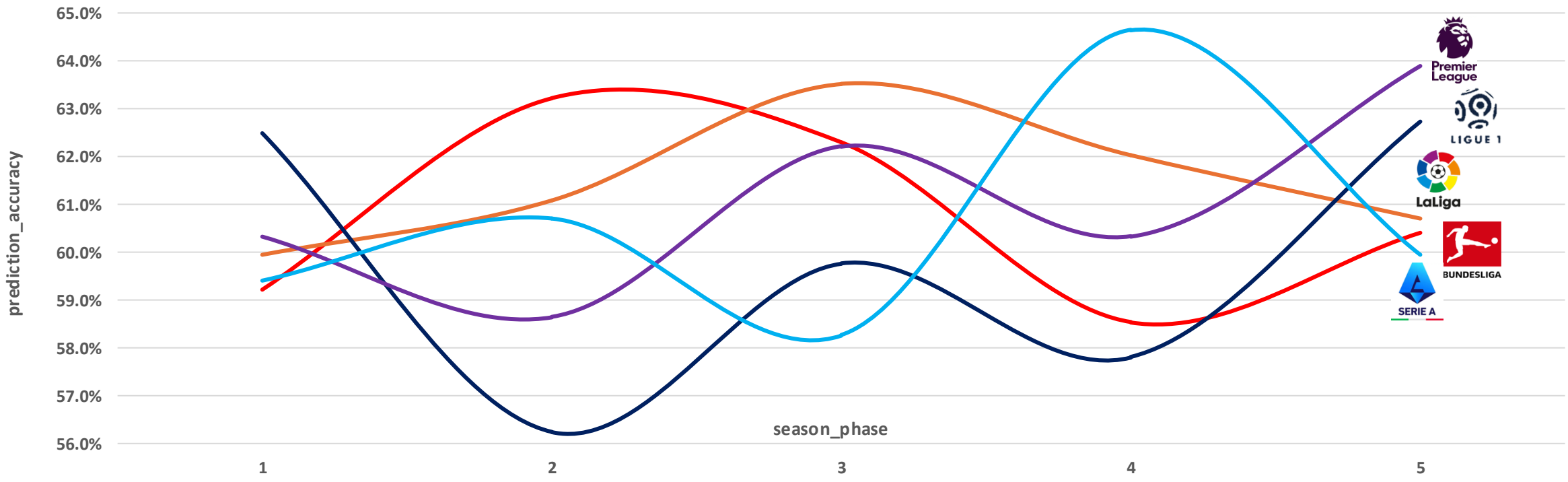
Asset Selection: Target clubs with repeatable, season-to-season attacking presence to secure dependable long-term brand exposure.
Risk Management: Use shorter, performance-linked, or flexible sponsorship structures for highly volatile teams.
Contractual Structuring: Stable assets justify multi-year commitments, whereas volatile assets require protective clauses against exposure swings.

Question 15: Seasonal Trends in Match Predictability

Maximize ROI by aligning campaign timing with seasonal predictability: riding early-season "chaos" for engagement and late-season "certainty" for reach.

Business Question: How does league predictability change over a season?
Commercial Rationale: Uncertainty levels dictate the emotional tone of a match, timing marketing spend allows brands to capture volatility-driven engagement early or peak audience attention during stable title races.

Predictability tends to rise as the season progresses across all leagues,
reflecting decreased squad uncertainty and the emergence of dominant title contenders .



Early-Phase Activation: Launch emotion-driven "upset" storytelling early to capture the high engagement driven by early-season unpredictability .
Late-Phase Premium: Concentrate investment on title run-ins when results are more "as expected" and audience attention is at its maximum.
Strategic Budgeting: Optimize ROI by utilizing early volatility for social media virality and late-season stability for guaranteed broadcast reach .

Conclusion: From Raw Matches to Sponsor-Ready Intelligence

1 What we have built

- **A two-layer relational DBMS** (raw → normalized core)
- **Governed schem**a with keys, constraints, lookup tables
- **Sponsor-oriented SQL layer** answering 15 commercial questions

2 Why it matters analytically

- Normalized per-90 / per-match metrics **make entities comparable**
- Volatility, predictability & involvement metrics **quantify exposure + risk**
- Enables **repeatable, auditable, statistically fair comparisons**

3 What it enables for brands

- **Choose where to invest** (league action + volatility maps)
- **Pick who to back** (Safe Stars, hidden-value players)
- **Decide how to structure deals** (multi-year vs flexible, risk-protected)

The system turns performance data into a **sponsor-oriented decision engine**: brands can now choose leagues, clubs and ambassadors based on transparent, normalised metrics - designing portfolios, campaign timing and contracts with evidence rather than intuition, and **laying the groundwork for future predictive and real-time sponsorship analytics.**

FIFA Premier League LaLiga LIGUE 1 BUNDESLIGA

THANK YOU!

Whether you're a brand manager maximizing screen time, an investor scouting the next big star, or just a hopeless optimist who keeps betting on Manchester United and Arsenal for Premier League Champion, our Analytics Platform is your strategic partner for turning blind faith into calculated, data-driven success.

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